

Trainable computation in molecular networks

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Reports of learning in single cells without genetic change span decades yet remain controversial, in part because there is no accepted general molecular mechanism for training comparable to gradient-based training or Hebbian learning in neural circuits. Here we identify a minimal set of ingredients sufficient to realize non-genetic learning, drawing inspiration from Boltzmann neural networks. First, dense reversible interaction networks provide an expressive substrate in which modulating the concentrations of a small set of mediator species can reprogram function without altering the underlying interaction parameters. Second, a simple rate-sensitive autoregulatory scheme that adjusts these mediator levels provides a local Hebbian-like training rule that can train the same network for diverse tasks, including Pavlovian conditioning, supervised classification, and generative tuning of bet-hedging ratios to match environmental statistics. We show that this autoregulatory training rule is model free and applies to reversible multimerization networks of arbitrary complexity, so training can compensate for unknown or unmodeled interactions present in vivo. These results suggest design principles for trainable synthetic cellular circuits and indicate how molecular systems could learn statistical features of their environments through experience.

Natural selection is often invoked to explain how cells come to embody the statistics of their past environments. In this view, regulatory programs encode those statistics, shaping cellular responses based on how frequently different conditions occurred in the past and not just on their immediate presence. These evolved programs can be remarkably sophisticated: microbes can anticipate nutrient depletion [1–4], populations hedge their bets by adopting diverse phenotypes at characteristic frequencies [5–9], and stress response pathways activate in anticipation of correlated environmental changes [10–12]. Here, memory of environmental statistics is genetically hardwired over evolutionary timescales through mutation and selection.

Yet many observations hint that cells can learn environmental statistics on much shorter timescales, via parameter remodeling within a lineage rather than differential survival of variants. These observations [8, 13–26] include habituation, priming, Pavlovian associative conditioning, anticipatory regulation, and learned bet hedging in cells ranging from bacteria to protists, yeast, and immune and cancer cells. While intriguing, many of these observations remain controversial, especially since the mechanisms underlying these phenomena are not fully clear.

Here, we define learning as experience-dependent reprogramming of regulatory programs: repeated exposure to a structured environment adjusts internal parameters so that future responses reflect correlations in past stimuli rather than only their immediate presence [27–29]; see Fig. 1a. Biologically, learning within a lifetime would allow cells and organisms to adapt to evolutionarily novel or fluctuating environments on short timescales and without the cost of deleterious mutations. In a synthetic biology context, it would enable the engineering of robust synthetic circuits that are trained *in situ* to perform useful behaviors, alleviating the need for mechanistic models

or repeated rounds of fine-tuning.

At present, no general mechanism is known by which molecular systems can learn from experience across diverse molecular architectures and learning behaviors. In comparison to learning in neural systems, learning in non-neural cells has seemed mechanistically implausible because of two fundamental obstacles. First, the relevant internal parameters, analogs of synaptic weights in neural networks, must be tunable without mutations, which rules out direct molecular affinities as potential training parameters. Second, a molecular ‘training rule’ must locally adjust these parameters to improve performance, without access to global computations such as back-propagation that enable learning in artificial neural networks.

Here, we show that both obstacles can be overcome in reversible molecular networks in which each species interacts with many others. In such densely interacting networks, the concentrations (or total abundances) of a small set of mediator, ‘hidden’ species can serve as trainable variables that are readily modulated by regulation without requiring mutations [28, 30–33]. Moreover, adapting training ideas from Boltzmann machines [34, 35] to the molecular domain, we show that simple rate-sensitive autoregulation of these hidden species yields a general training rule that tunes their concentrations toward values consistent with environmental correlations experienced during training; see Fig. 1b. Across different environmental contexts, this rate-sensitive autoregulatory rule enables a wide class of reversible molecular networks to undergo Pavlovian conditioning, learn quantitative input–output relationships, perform supervised classification, and learn generative behaviors relevant for bet hedging. We argue that the required molecular ingredients, dense interaction networks, autoregulation, and rate sensitivity, are widespread in biology. Our findings

therefore suggest a broadly applicable, model-free mechanism by which molecular systems can reprogram their function on physiological timescales without requiring genetic change.

I. TRAINING FOR PAVLOVIAN CONDITIONING

We begin with the classic problem of Pavlovian conditioning [14, 36], in which a neutral stimulus becomes predictive of a biologically potent one. Pavlovian conditioning in non-neural contexts have been claimed since Gelber’s early work [37] on protists, with subsequent observations in ciliates and amoebae, slime molds and plants, and, more recently, in macrophages and in cancer drug resistance [17, 38–47]. However, alternative explanations and the lack of molecular mechanisms render many of these claims controversial [13, 22]. Prior work has explored diverse molecular routes to learning and memory, ranging from evolutionary and phenomenological proposals to concrete task-specific circuits [1, 2, 28, 29, 48–59]. Here we develop a physically grounded framework for Pavlovian conditioning in molecular systems inspired by Boltzmann-machine training [34, 35, 60] (SI Sec. I). Related ideas have been adapted to engineered mechanical and electrical systems [61–71]. Our approach here applies to a broad class of reversible molecular networks, assuming only that environmental stimuli can shift their steady state (SI Sec. II). Within this setting, we show how a simple autoregulatory feedback rule enables associative learning at the molecular level. Extensions to supervised and unsupervised learning are described later.

In our molecular Pavlovian setup, an environmental signal E_B acts as the *neutral stimulus* (“bell”) and another E_F as the *potent stimulus* (“food”) (Fig. 2). E_B modulates an internal species V_B , while E_F affects a response species V_F . For concreteness, we take the monomeric form V_F^{mon} as the functional response, although variants where V_F^{tot} or V_F^{dim} serve as response behave equivalently (SI Sec. IV). Initially, E_B alone produces no consistent V_F response. Training seeks to modify internal molecular states so that, after repeated co-presentation of E_B and E_F , the neutral stimulus alone elicits the response previously induced by the potent one.

To formalize this, we adopt the language of Boltzmann-machine training: molecular species directly coupled to the environment (V_B , V_F) are *visible*, and those coupled only indirectly are *hidden* (H_i , Fig. 2a). Visible and hidden species reversibly form complexes like dimers, trimers, etc. through affinities K_{ij} . Unlike neural networks that tune coupling weights, our model treats the total concentrations of hidden species H_i^{tot} as trainable variables, since concentrations can change dynamically within a lifetime whereas affinities K_{ij} are genetically fixed. The training rule introduced below operates without knowledge of K_{ij} and is compatible with many network topologies (SI Sec. II).

Our proposed general training rule, derived by adapting Boltzmann learning ideas to molecular systems (SI Secs. I, V), says that each hidden species H_i must autoregulate based on rapid changes in its own monomer levels H_i^{mon} : fast increases or decreases of H_i^{mon} must trigger production or degradation while steady levels of H_i^{mon} must leave H_i^{tot} unchanged. Mathematically

$$\frac{dH_i^{\text{tot}}}{dt} = -F_{HP}[H_i^{\text{mon}}(t)], \quad (1)$$

where F_{HP} represents a high-pass filter with filter timescale τ_{HP} . Many biophysical and biomolecular mechanisms can naturally lead to high-pass filtering; these include classically studied circuit motifs such as integral feedback [72] (e.g., bacterial chemotaxis [73–75], fungal osmotic regulation [76–78]), incoherent feed-forward loops [79] (e.g., NF κ B signaling [80–82], ERK pathway [83, 84] and dictyostelium chemotaxis [85]) and fold-change detection circuits [86–88] and pathways with paradoxical signaling (or end product inhibition) [89, 90]. In addition, high pass filtering is also implicit in the natural biophysics of cellular phenomena such as receptor desensitization [91–94], resource depletion [95–97], phase separation [98–100] and other buffering mechanisms. See SI Sec. V for further discussion of mechanistic realizations.

To illustrate the learning process, we consider a structured environment where the ‘food’ E_F appears in brief spikes (Fig. 2b) when the ‘bell’ E_B is high but not otherwise. Each spike drives the system between “sleep” (bell only) and “wake” (bell and food) equilibria. Updates to H_i^{tot} occur mainly during the rapid sleep-to-wake transition, when monomer concentrations shift rapidly on a timescale τ_f ; during the slow return to sleep over a period τ_s , changes are negligible.

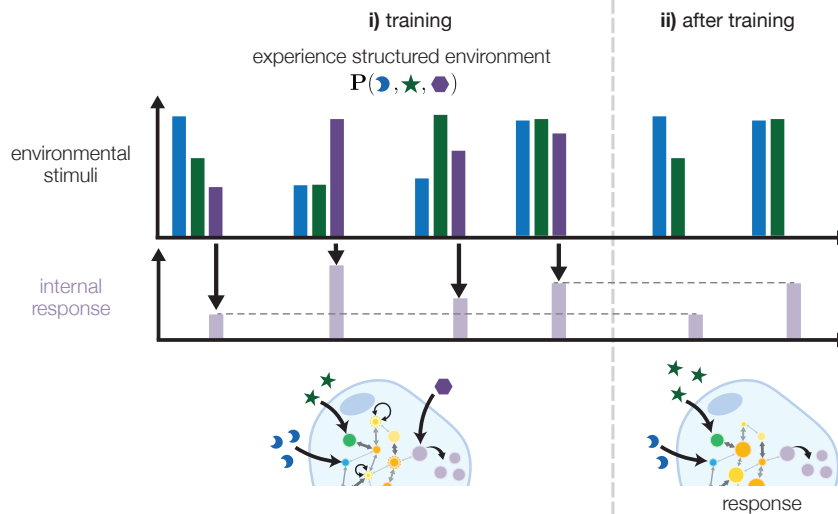
Thus, the training rule of Eq. 1 acts like friction: when $[H_i^{\text{mon}}]$ changes quickly, the update to $[H_i^{\text{tot}}]$ opposes that change, so the most strongly fluctuating species are damped most. Accumulated over repeated correlated inputs, these frictional updates drive the network toward hidden-species totals that make the sleep and wake equilibria similar, thereby encoding the learned association.

We demonstrate our proposed mechanism in a minimal three-species dimerization network (Fig. 2b) comprising visible species V_B , V_F , and a hidden species H that interact through fixed binding affinities K_{ij} .

For an initially high $[H^{\text{tot}}]$, the sleep state (only E_B present) yields a too-low output $[V_F^{\text{mon}}]$, because H binds strongly to V_F , forming dimers that deplete V_F monomers. During the wake phase, spikes in E_F increase V_F^{tot} to soak up enough H^{mon} so that $[V_F^{\text{mon}}]$ increases to the right level. The lower $[H^{\text{mon}}]$ during wake serves as an error signal: the training rule decreases H^{tot} to form fewer V_F - H dimers, thereby counteracting the excessive sequestration of V_F^{mon} .

Through repeated cycles, this feedback minimizes the difference in H^{mon} between sleep and wake by decreasing $[H^{\text{tot}}]$. Once the distribution of each species across

a) Single cell learning from environmental correlations



b) Minimal ingredients for learning

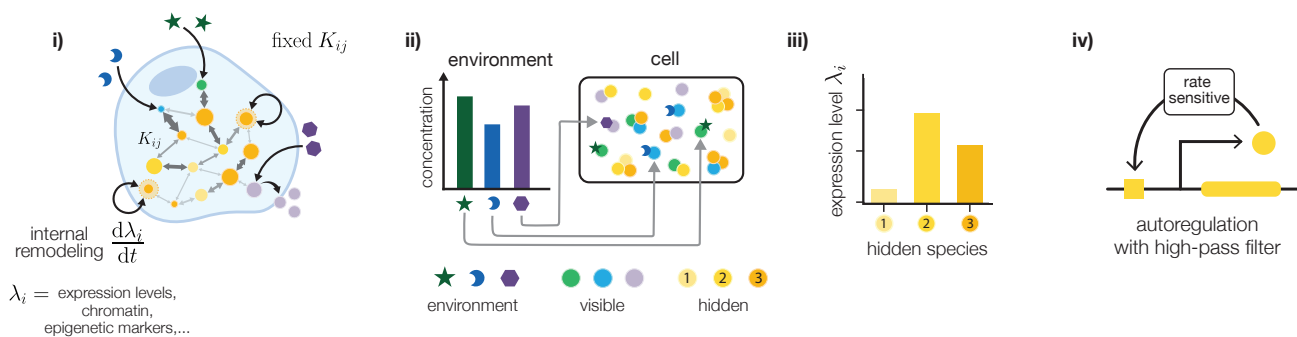


FIG. 1. Trainable molecular networks enable single cells to learn environmental correlations. (a) Operational definition of learning. Bars show the levels of several environmental inputs (blue, green, purple) that fluctuate jointly according to a structured distribution $P(\Delta, \star, \circ)$. We focus on an internal response (light purple) induced directly by the purple stimulus (hexagon). During exposure to a structured environment (i), a cell repeatedly experiences correlated combinations of stimuli that drives changes in its internal molecular parameters. After training (ii), blue and green inputs by themselves elicit an internal response that approximates the response that would occur if the purple cue were present; such a molecular network is said to have learned the statistical structure of environmental stimuli. (b) Minimal molecular ingredients for learning: (i) We consider a dense reversible interaction network between many molecular species with fixed microscopic interaction parameters K_{ij} . Training acts by slowly tuning other internal parameters λ_i according to a training rule while leaving K_{ij} unchanged. (ii) Environmental molecules impact a subset of internal “visible” species, shifting the equilibrium of the internal network. Other internal components, “hidden” species, do not couple directly to the environment but influence cellular response. (iii) In our study, concentrations of hidden species play the role of training parameters λ_i . (iv) Training rule: Hidden species levels are trained by a rate-sensitive autoregulatory rule: rapid changes in the concentration of a monomer drives production or degradation of that species, whereas steady monomer levels do not (i.e., autoregulation with high-pass filtering).

monomers and dimers is similar in both states, the output $[V_F^{\text{mon}}]$ becomes invariant between them. The neutral stimulus E_B alone now elicits the same response previously requiring E_F . Correspondingly, the deviation of $[V_F^{\text{mon}}]$ from its target value decreases to near zero over training epochs (Fig. 2b, right), while $[H^{\text{tot}}]$ decreases by almost an order of magnitude, encoding the learned association in steady state.

The success of training above does depend on a hierarchy of timescales; we require that the equilibration time

τ_{eq} of the reversible molecular network (e.g., for multimerization) is faster than the environmental timescale τ_f which in turn is faster than τ_s , with filtering timescale τ_{HP} of autoregulation (Eq. 1) being intermediate to τ_f and τ_s . See SI Sec. V for further discussion of timescales.

This adaptive feedback mechanism is broadly applicable: as shown later, any reversible molecular network whose equilibrium distribution shifts with environmental inputs can, in principle, be trained by the same rule. The expressive capacity of such systems (i.e., range of be-

haviors such networks can feasibly achieve), determined by the underlying affinity matrix K_{ij} , has been analyzed elsewhere [101–104] and is conceptually distinct from the question of how such networks can learn *in situ* studied here. Following these prior analyses, we assume dense all-to-all couplings in which each species interacts with many others, providing sufficient expressivity for the behaviors studied below.

II. SUPERVISED TRAINING FOR INPUT-OUTPUT BEHAVIORS

The same framework extends naturally beyond binary associations to quantitative input–output mappings between levels of E_B and V_F . No modification of the autoregulatory rule is required; the only new ingredient is that, during training, the potent stimulus E_F spikes with amplitudes that depend on the level of E_B .

For example, when low and high levels of E_B coincide with small and large E_F spikes, respectively, the depicted three-species dimerization circuit learns a step-down relationship between E_B and V_F . Similarly, an eight-species network learns a non-monotonic (low–high–low) mapping if the E_F spike amplitudes correlate accordingly with E_B , and can also implement a two-input function when exposed to two input signals, E_1 and E_2 , alongside the spiking E_F (Fig. 3c). Implementation details are provided in SI Sec. VI.

To test our framework’s learning capacity on more complex problems, we applied it to the standard four-feature Iris classification task, treating the four feature values as four environmental signals $E_1, \dots, E_4$. After training on 80 points from two classes, the output species reliably classified the *setosa* and *versicolor* flower types with $\sim 89\%$ training and $\sim 90\%$ test accuracy (Fig. 3d), demonstrating that the same molecular rule can scale to higher-dimensional inputs and support generalization beyond simple associative tasks.

III. TRAINING FOR A DISTRIBUTION OF PHENOTYPES

The same autoregulatory training rule also extends to circuits whose behavior is stochastic and distributional rather than a deterministic input–output map. Many biological systems exhibit stochastic phenotypic variability [5], in which genetically identical cells occupy distinct states at characteristic frequencies and switch between them over time. Such controlled variability supports bet hedging across diverse settings, including bacterial persistence [105], yeast stress-activation heterogeneity [7], developmental fate diversification [106, 107], phage lysis–lysogeny [108], and bacterial competence [109, 110].

The occupancy fractions of different phenotypes are generally assumed to be genetically hardwired by evolution to match long-term environmental frequencies

[6]; these fractions have also been hardwired in synthetic circuits [111, 112]. However, several studies [7–9, 18, 23, 24, 113] suggest that such fractions can instead be learned from experience, but a general molecular mechanism for tuning phenotypic distributions remains unknown. Here we show that our rate-sensitive autoregulatory rule provides such a mechanism.

Bistable molecular circuits provide a natural substrate for generating distributions of phenotypes, since intrinsic noise drives transitions between two attractor states. In many natural and synthetic examples (Fig. 4), the occupancy of these states can be tuned by adjusting the level of a specific molecular species. This suggests a simple principle: regulating the concentration of such a species with a training rule can train the stationary distribution.

We test this idea using a minimal bistable network composed of two mutually inhibitory species (A , B), whose relative balance is controlled by a hidden species H that sequesters B into an inactive complex (Fig. 4a). Increasing $[H^{\text{tot}}]$ stabilizes the high- A /low- B state, while decreasing it shifts the balance towards the high- B state. Although this probabilistic behavior differs qualitatively from associative or supervised learning, SI Sec. II shows that the Boltzmann-learning framework predicts that autoregulation based on $[H^{\text{mon}}]$ will tune the stationary occupancy of the two attractors to match experienced environmental frequencies.

In simulations, we initialize the switch with an occupancy bias (i.e., level of $[H^{\text{tot}}]$) that differs from the environment (e.g., 50/50 vs. 10/90 for the low- and high- A states; SI Sec. II). During training, an environmental signal E_A drives the system into the two configurations with frequencies $P^{\text{env}}([A]^{\text{tot}})$ that reflect environmental statistics, inducing fluctuations in H^{mon} that in turn autoregulate H^{tot} . After repeated training cycles, $[H^{\text{tot}}]$ converges to a value that spontaneously reproduces the environmental occupancy fractions experienced in the past, even in the absence of any environmental signal E_A (Fig. 4b). In a cellular population, this corresponds to a learned allocation strategy in which the fraction of cells adopting each phenotype reflects the statistics of previously encountered environments.

IV. THE TRAINING FRAMEWORK IS MODEL-AGNOSTIC

Thus far, we have demonstrated our auto-regulatory training framework on dimerization networks and bistable switches. Since this rule was inspired by Boltzmann machine training, a natural question arises: can we derive training rules that work for more complex molecular systems? Many biological networks involve reversible associations beyond simple dimerization, including trimerization, higher-order complexes, or even liquid condensates. Would each such molecular system require a distinct training rule, or could the same training mechanism apply broadly?

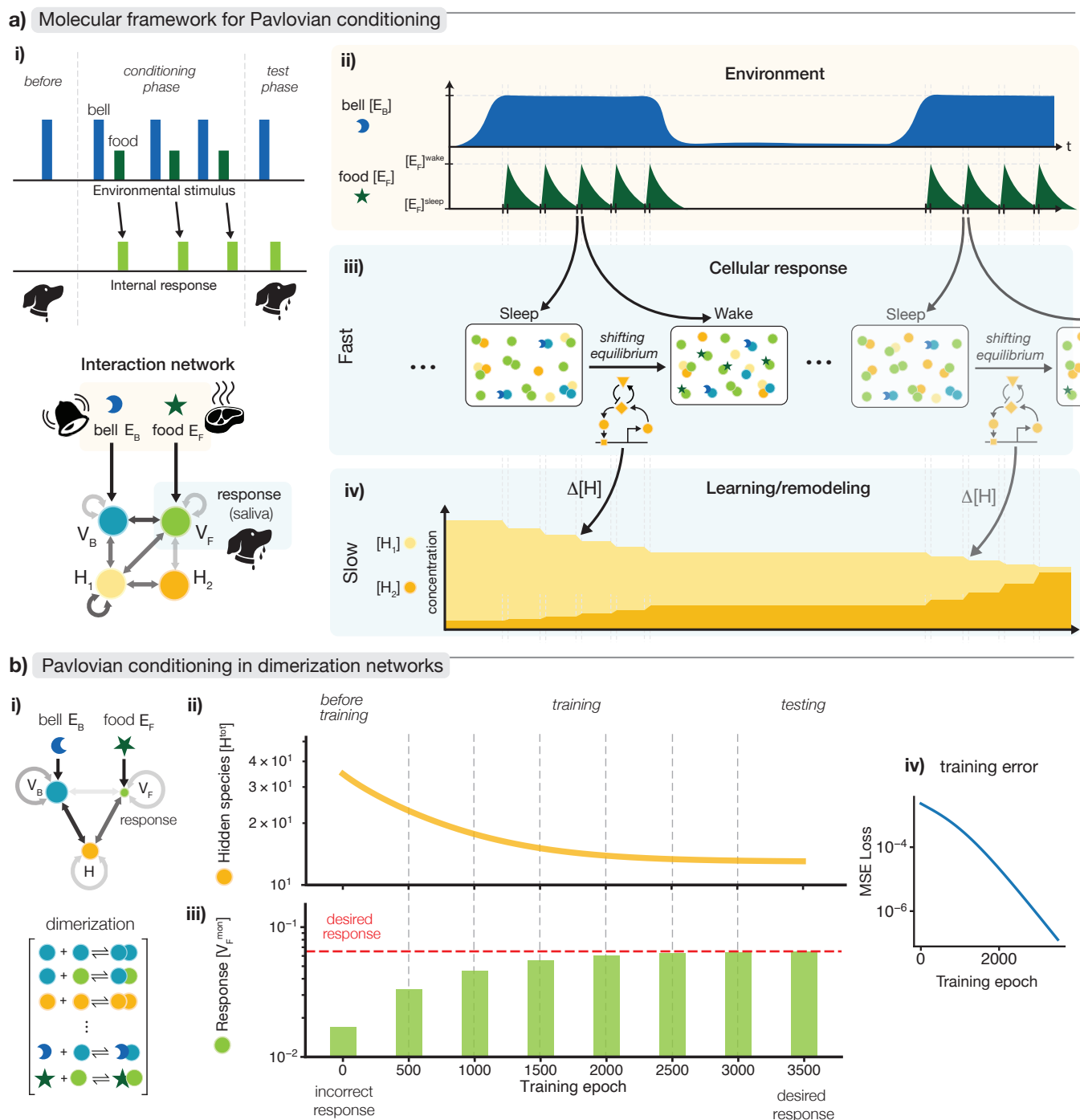
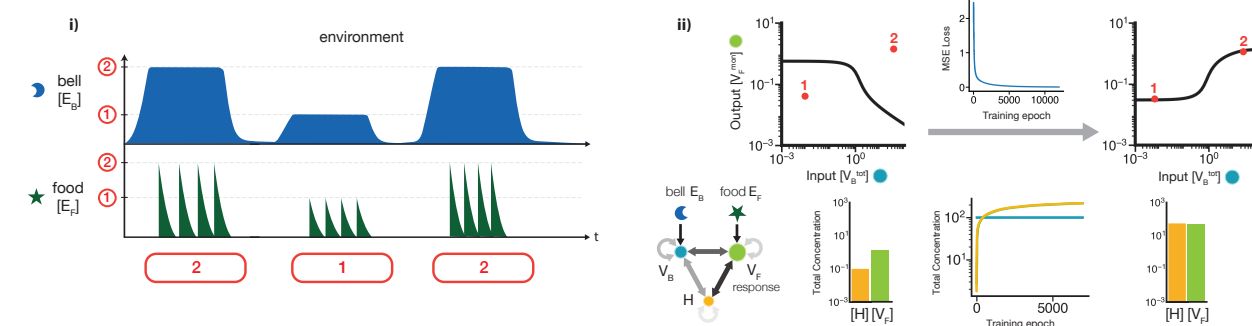
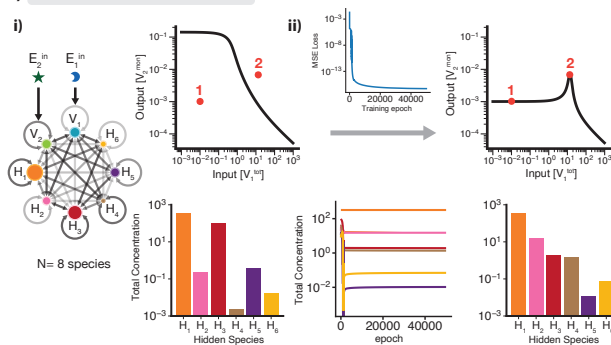


FIG. 2. Pavlovian conditioning through autoregulation of hidden species in a structured environment. a. (i) Schematic of molecular Pavlovian conditioning. (ii) The neutral stimulus (*bell*, blue) and potent stimulus (*food*, green) correspond to environmental molecules E_B and E_F , which modulate internal visible species (V_B , V_F) within a network that also contains hidden species (H_1 , H_2). (iii) Training is induced by temporally correlated inputs (blue and green curves, upper right), where the presence of the bell molecule E_B coincides with transient spikes in the food molecule E_F . These spikes rapidly shift the network from a “sleep” equilibrium with low V_F^{mon} to a “wake” equilibrium with high V_F^{mon} and then slowly back to sleep. (iv) On a slower timescale, autoregulatory feedback adjusts the total levels H_1^{tot} and H_2^{tot} in those species whose monomer levels fluctuate most rapidly, so as to reduce those fluctuations. As training converges, hidden-species levels adjust so that E_B alone elicits a high V_F response. **b.** (i) Pavlovian conditioning in a three-species dimerization network (V_1 , V_2 , H) coupled to two stimuli E_F , E_B . Grayscale values of network edges represent strengths of binding constants K_{ij} which stay fixed during training. (ii) Accumulation of the hidden species H in response to spikes of E_F that occur only when E_B is high during training. (iii) Testing during training: output response [V_F^{mon}] to periodic test pulses of the neutral-stimulus molecule E_B presented without E_F . As learning proceeds, these test responses increase toward the desired level. (iv) Mean-squared error (MSE) between actual and target output V_F^{mon} in response to E_B decreases over training epochs, demonstrating Pavlovian conditioning.

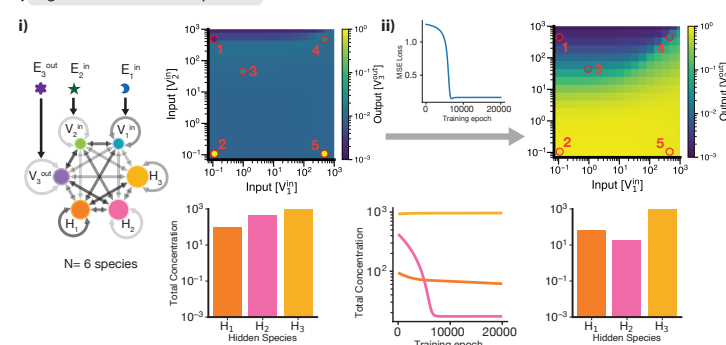
a) Molecular framework for supervised learning



b) Monotonic to non-monotonic



c) Higher dimensional responses



d) Classification of Iris dataset

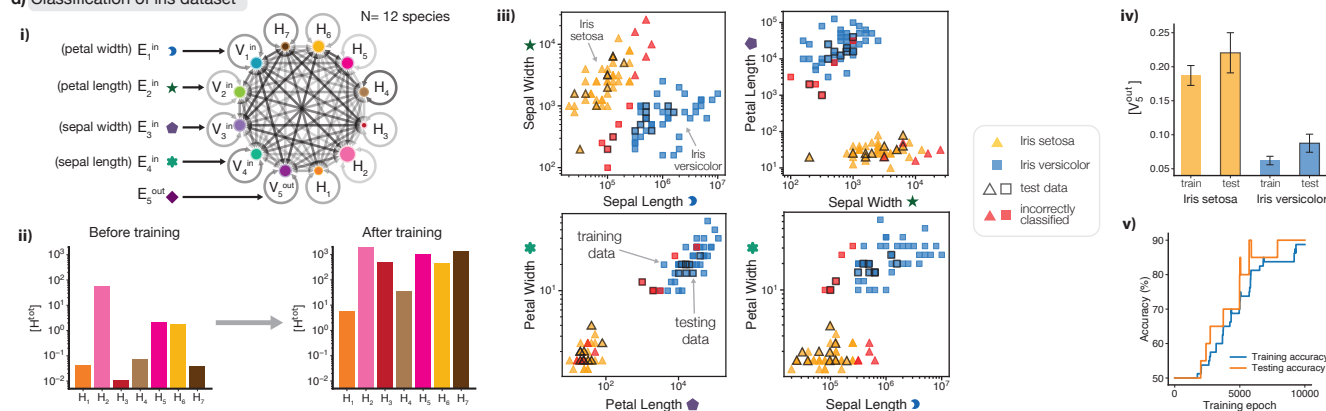
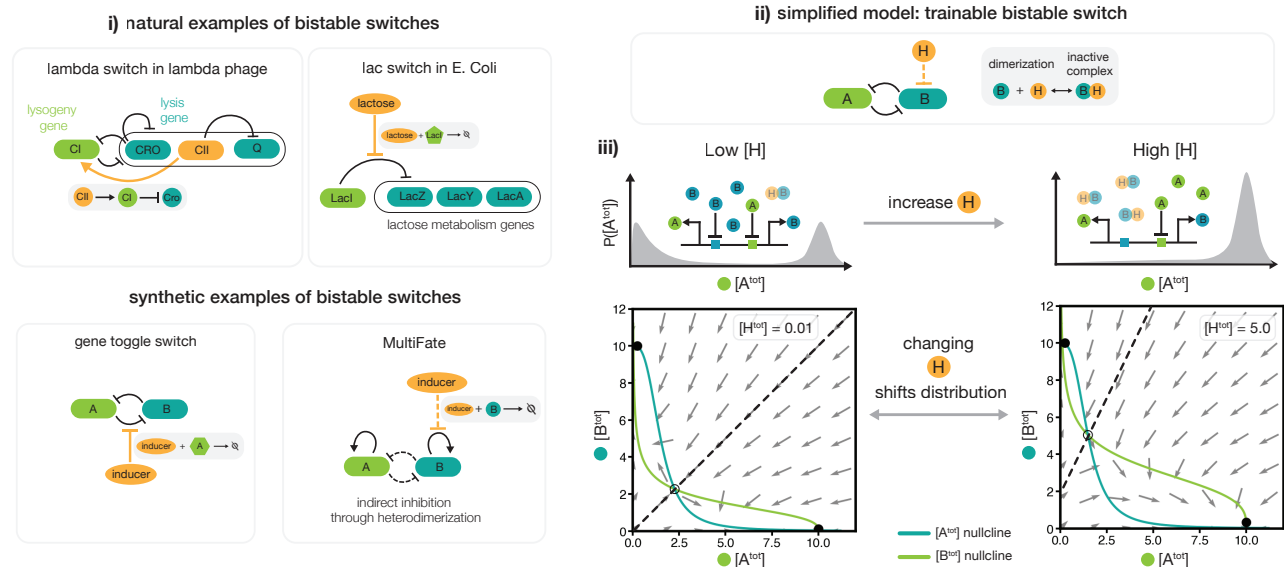
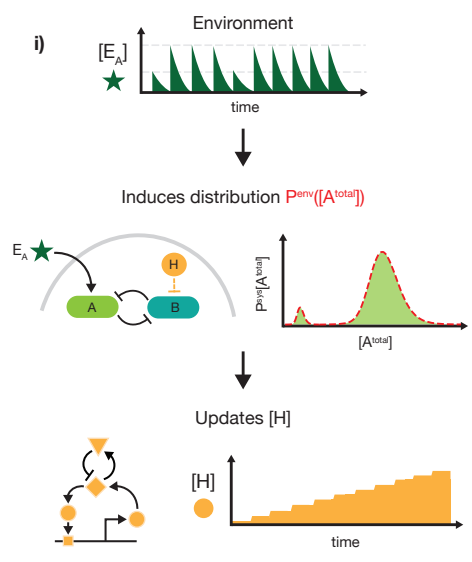


FIG. 3. Supervised molecular learning of quantitative and multidimensional input–output behaviors. **a.** (i) Schematic of the training environment for supervised learning. Environmental inputs E_B (“bell”) and E_F (“food”) vary in correlated patterns such that high E_B coincides with high amplitude E_F spikes (training point 1) and vice versa (training point 2). (ii) A three-species network (V_B , V_F , H) initially implements an incorrect input–output relation between $[V_B^{tot}]$ and $[V_F^{mon}]$. During training, hidden-species levels adjust themselves through the auto-regulation training rule (Eq. 1), reducing deviation from the target mapping. After training, the network reproduces the environmental correlations: high E_B yields high V_F , and low E_B yields low V_F . **b.** Extension to more complex mappings. (i) An eight-species network learns a non-monotonic response of output $[V_2^{mon}]$ to input $[V_1^{tot}]$. (ii) Internal concentrations evolve and the mean-squared error (MSE) from the target output decreases over training. **c.** The same rule generalizes to multidimensional inputs: (i) an eight-species network learns a 2D input–output function mapping ($[V_1^{tot}, V_2^{tot}]$) to $[V_3^{mon}]$. (ii) Training data shown in red circles. **d.** Classification of the Iris dataset using a twelve-species network. (i) Four input features (sepal and petal length and width) are encoded in stimuli (E_1, E_2, E_3, E_4) which induce concentrations of visible species ($[V_1^{tot}], [V_2^{tot}], [V_3^{tot}], [V_4^{tot}]$), and the output molecule $[V_5^{mon}]$ represents flower identity (low for *I. setosa*, high for *I. versicolor*). After training, (ii) the network reorganizes hidden species concentrations to (iii) separate species (triangle is *setosa*, square is *versicolor*, and red points are classified as the incorrect species). Training results in high $[V_5^{out}]$ for *setosa* and low $[V_5^{out}]$ for *versicolor*, classifying points with (iv) ~89% accuracy on the training set and ~90% accuracy on the test set and demonstrating that the same molecular training rule can generalize to high-dimensional classification tasks.

a) Trainable multistable molecular systems



b) Molecular framework for generative learning



c) Training a bistable switch as a generative model

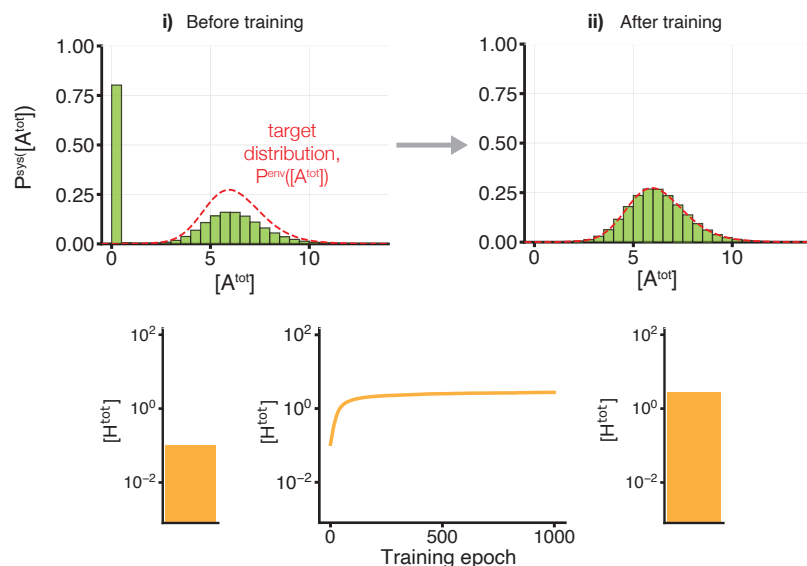


FIG. 4. Training phenotype occupancy of a noisy bistable switch to match environmental statistics. **a.** (i) Examples of natural [114, 115] and synthetic [116, 117] bistable molecular switches in which the concentration of one molecule (orange) tunes the relative stability of two attractor states. (ii) A trainable bistable model composed of two mutually inhibiting species (A, B); a hidden species H sequesters B into an inactive complex. The two stable states correspond to high- A /low- B and low- A /high- B configurations. (iii) Probability distributions $P^{sys}([A^{tot}])$ in the presence of intrinsic noise and phase-plane plots for low and high $[H^{tot}]$; changing $[H^{tot}]$ shifts the relative stability and thus occupancy of these states. **b.** (i) During training, an environmental signal E_A drives the system into the high and low A states at frequencies $P^{env}([A^{tot}])$ (dashed red line; schematic) defined by the environment. The autoregulatory training rule adjusts the hidden species $[H^{tot}]$ (orange; schematic) based on changes in $[H^{mon}]$. **c.** (i) Environmental distribution $P^{env}([A^{tot}])$ (red) and the noisy bistable switch's distribution $P^{sys}([A^{tot}])$ (green), with initial untrained value of H^{tot} . Training of H was carried out as illustrated in (b). (ii) After training, the system's state-occupancy distribution $P^{sys}([A^{tot}])$ matches the environmental distribution $P^{env}([A^{tot}])$; levels of hidden species H change by $\sim 10\times$ during training.

In this section, we show that the training framework is model-agnostic: the training rule has the same functional form across reversible dilute association networks and relies only on measured concentration signals (and their transients). As a result, the training rule can be applied without knowing the physics of the underlying network (e.g., dimerization vs multimerization), interaction parameters, and remains effective under unmodeled interactions and cell-to-cell variability. In SI Sec. II, we derive the training rule for molecular networks that form general higher-order complexes. We find that the same auto-regulation rule applies across interaction complexity, provided that (1) interactions remain reversible and (2) complexes remain dilute rather than forming dense liquid phases. For liquid-liquid phase separation, the rule may still drive learning but is not mathematically guaranteed to be optimal; we leave this case to future work.

Figure 5a demonstrates this generality by training two 10-species networks, one with only dimerization and one including trimerization. Equilibrium constants (K_{ij} and K_{ijk}) were independently sampled from a log-normal distribution, and both networks were trained on the same input-output examples (red circles). Despite distinct interaction types, both networks learned the same input-output response, with the same training rule producing different trained values of $[H^{\text{tot}}]$.

This model-agnostic property also addresses a central challenge in synthetic biology: the ‘reality gap’ between design and implementation. Synthetic circuits are typically designed from explicit or implicit models of molecular interactions, yet *in vivo* behavior can deviate from *in vitro* or *in silico* predictions due to unmodeled cellular interactions, forcing iterative redesign[118, 119].

Our feedback-based training framework bypasses this dependence on accurate models. In Fig. 5b, a four-species network is first designed *in silico* by choosing K_{ij} and $[H^{\text{tot}}]$ to match a target input-output response, but adding unmodeled interactions with additional cellular species (red stars) disrupts the behavior. Training the hidden species H_1, H_2 via auto-regulation nevertheless restores the target response, without manipulating the interfering species (red stars) and without modifying the training rule to account for them.

Figure 5c extends this feature to heterogeneous populations. We simulated a population of 50 cells with 10-species networks in which each cell’s binding constants K_{ij} vary across orders of magnitude. Before training, a shared $[H^{\text{tot}}]$ produces diverse responses across cells; after exposure to the same training protocol, the auto-regulatory rule drives each cell to a different $[H^{\text{tot}}]$ that compensates for its specific K_{ij} . Consequently, all cells achieve the same target output (red dashed line), showing that a single training environment can program diverse molecular implementations to a common function.

V. THE TRAINING FRAMEWORK IS ROBUST TO IMPLEMENTATION DETAILS

The model-agnostic advantages above come with an apparent overhead: unlike static circuit design, training requires a plasticity mechanism (here, autoregulation of hidden-species levels) and an interaction protocol with the environment during a training period. A key practical question is how demanding those costs are in implementation. If successful training required a finely tuned, chemically specific realization of the training rule or a highly controlled training protocol, the added complexity could negate its benefits. Here we show that this is not the case: the framework’s performance is insensitive to many details of how training is realized, suggesting that the additional engineering burden can be modest.

First, training does not require the environment to clamp the output to its exact target. Figure 5e (top) shows that a simple “nudging” protocol, inspired by equilibrium propagation [68] and related physical implementations[66, 120, 121], is sufficient: after observing the output, the environment applies a small corrective push in the appropriate direction. Second, training can be discretized into separate sleep and wake phases with updates applied once per phase rather than continuously (Figure 5e, bottom). Although this slows learning, it relaxes the need for rate-based auto-regulation and still converges to the desired behavior (see SI for more).

Finally, the precise functional form of autoregulation is flexible. While the Boltzmann-machine derivation (SI Sec. I) yields an optimal update $\Delta[H^{\text{tot}}] \propto \log[H^{\text{mon}}]$, Figure 5d shows that molecularly plausible alternatives also train successfully, including linear responses (used throughout this paper) that might model post-translational activation of hidden species and Hill-type nonlinearities typical of transcriptional feedback. These variants primarily affect convergence speed rather than whether learning succeeds.

Together, these results indicate that the framework’s benefits do not hinge on a narrow mechanistic implementation, and that the extra overhead of training can be realized with coarse, biophysically natural approximations rather than finely tuned control.

VI. DISCUSSION

We showed that dense, reversible molecular interaction networks form an expressive computational substrate in which the concentrations of a few hidden species act as trainable weights, analogous to synaptic weights in neural networks. Training does not require backpropagation but instead follows from rate-sensitive autoregulation of hidden species. This mechanism supports Pavlovian conditioning, supervised classification, and tuning of stochastic phenotypic fractions, allowing molecular systems to acquire new regulatory functions within a lifetime without changing microscopic affinities. More

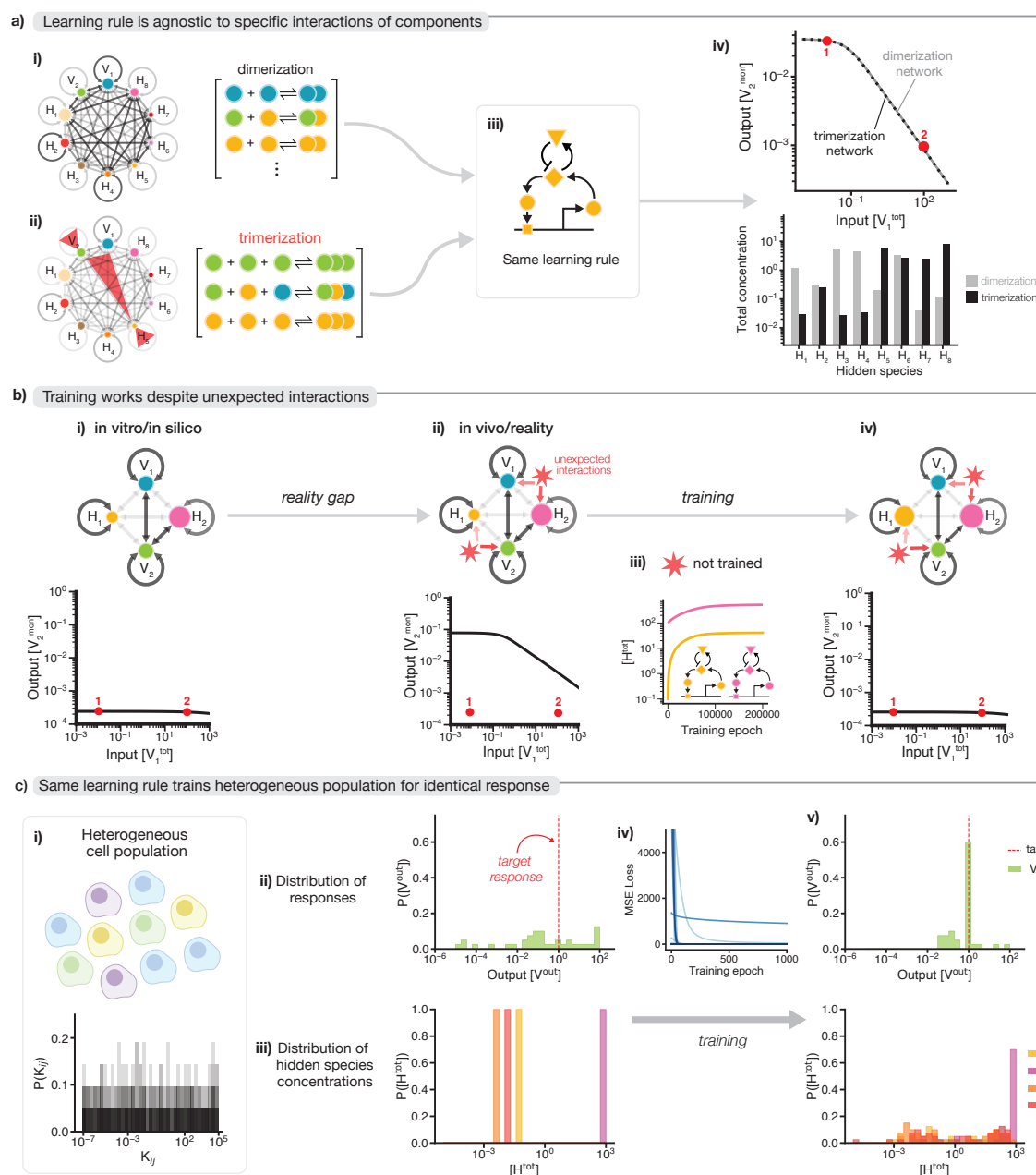


FIG. 5. Auto-regulatory training is model-agnostic and hence robust to unmodeled interactions and population heterogeneity. **a.** A ten-species network trained with either (i) trimerization or (ii) dimerization interactions. The same auto-regulatory training rule successfully trains both networks to identical input-output behavior using the same environmental examples, even though this requires tuning $[H^{\text{tot}}]$ to different values in the two networks. **b.** (i) A four-species network optimized *in silico* for a specific behavior produces (ii) incorrect behavior when unmodeled interactions with new interfering species (red stars) are introduced. (iii) Training via autoregulation on H_1 and H_2 alone recovers the desired input-output map without requiring knowledge, training or manipulation of the unmodeled species (red stars). **c.** (i) A heterogeneous population of cells, each with a 10-species network whose binding constants K_{ij} are randomly sampled from a wide distribution. (ii, iii) Initially, input-output behavior and hidden species concentrations vary widely across the population. (iv, v) After training in the same structured environment, each cell converges to different $[H^{\text{tot}}]$ values that compensate for that cell's idiosyncratic K_{ij} , so that all cells yield the same target response (red dashed line).

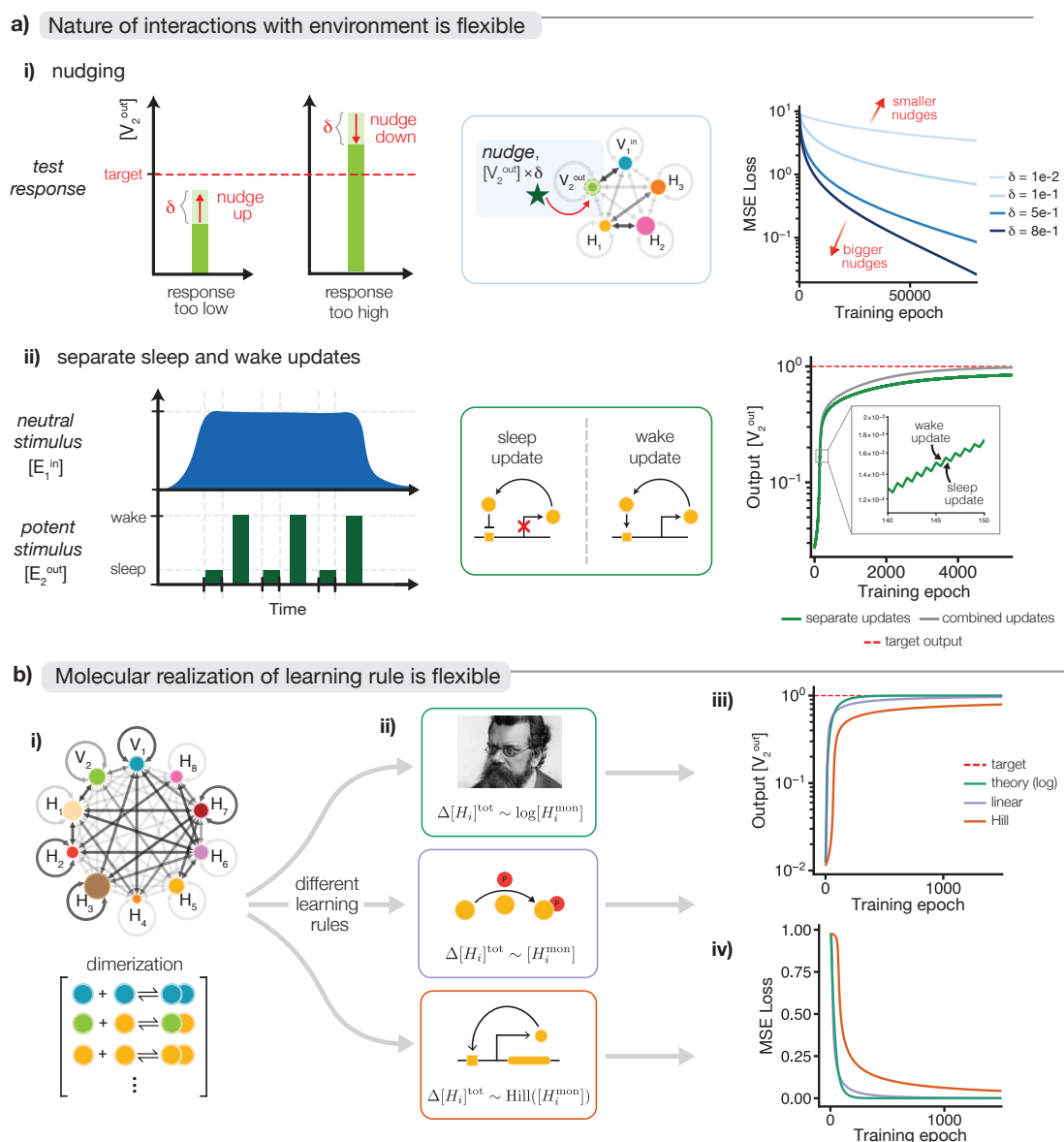


FIG. 6. Robustness of the training framework to implementation details. **a.** Flexibility in training protocols: the training rule works with alternative models of environmental interaction. (i) The spiking environmental signal (food E_2^{out}) need only nudge the response molecule V_2^{out} toward the desired level rather than imposing it exactly. A five-species network trained with nudging signals successfully decreases MSE loss, with larger nudges resulting in faster convergence. (ii) Separate sleep and wake phase auto-regulatory updates (with opposite signs) alleviate the need for rate-based auto-regulation. Output V_2^{out} over sleep wake cycles converges to the target level (dashed red), but exhibits oscillations (inset) and larger final error relative to combined updates. **b.** (i, ii) Different mathematical forms of the auto-regulatory training rule (logarithmic, Hill, or linear proportional) applied to train a 10-species dimer network. (iii, iv) The mathematically optimal rule $\Delta[H^{tot}] \propto \log[H^{mon}]$, derived from Boltzmann machine theory, converges fastest, but molecularly realistic rules based on post-transcriptional activation (purple) or transcriptional auto-regulation (orange) also successfully evolve toward the desired behavior.

broadly, this work demonstrates that physical systems can naturally embody learning algorithms typically associated with neural networks, in line with learning behavior in electrical, mechanical or mechanical-metamaterial networks[61–71, 120, 122–127]

The key ingredients needed – promiscuous interac-

tion networks, autoregulation of expression or activity, and rate-sensitive signaling that distinguishes transient from persistent inputs – are widespread in biology. Thus cells that experience structured environments could exploit such mechanisms to tune their responses to environmental statistics on timescales much shorter than

evolution. Our work suggests specific experimental signatures for molecular implementations of learning, e.g., species whose concentrations change most strongly during a training period are predicted to be rate responsive and autoregulatory, and to interact promiscuously with many other molecules.

Our results also suggest a new, model free paradigm for synthetic biology. Traditional circuit design relies on detailed mechanistic models and repeated rounds of redesign when *in vivo* implementations deviate from expectations. By contrast, a trainable circuit can be assembled from expressive but imperfect components and then trained directly *in situ*[56, 57], so that internal molecular weights self adjust to realize a desired function despite unknown interactions. One can envisage, for example, an engineered immune cell whose synthetic circuit is trained on panels of healthy and cancer cells until it reliably triggers an effector program on the correct targets, thereby exploiting real correlations in the training environment rather than a predefined model. The training framework is intrinsically robust to inevitable unmodeled interactions found *in vivo*, because the training rule never assumes a specific model of the reversible network that it trains.

Our analysis has important limitations that suggest directions for future work. The auto-regulatory training rule was derived for reversible binding and assumes a separation of timescales between molecular equilibration, autoregulation, and environmental variation, and was limited to static input-output tasks. As a consequence, dense condensates[60] and complex temporal sequence processing[29] fall outside the present analysis.

Another key challenge involves long term inheritance of learned behavior across many cell divisions. Persistent memory of learned hidden-species expression levels is likely to require multistable attractors for those species, epigenetic memory[128, 129], or continual reinforcement by the environment. Future work should probe how evolution and learning processes might interact and sculpt trainable networks, for example by evolving network architectures under selection for rapid, flexible trainability across diverse tasks within a lifetime.

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